Algebra of Functions

If f and g are functions, and x is an element of the domain of each function, then

$$(f+g)(x) = f(x) + g(x)$$

$$(f-g)(x) = f(x) - g(x)$$

$$(f \bullet g)(x) = f(x) \bullet g(x)$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, g(x) \neq 0$$

Let : $f(x) = x^2 + 4x + 3$

$g(x) = x + 3$

Find: $(f + g)(2)$

$$= (x^2 + 4x + 3) + (x + 3)$$

$$= x^2 + 4x + 3 + x + 3$$

$$= x^2 + 5x + 6$$

$$= 2^2 + 5(2) + 6$$

$$= 20$$

Let : $f(x) = x^2 + 4x + 3$

$g(x) = x + 3$

Find: $(f - g)(1)$

$= (x^2 + 4x + 3) - (x + 3)$

$= x^2 + 4x + 3 - x - 3$

$= x^2 + 3x$

$= 1^2 + 3(1)$

$= 4$

Let : $f(x) = x^2 + 4x + 3$

$g(x) = x + 3$

Find: $(f \bullet g)(-1)$

$= (x^2 + 4x + 3) \bullet (x + 3)$

$= x^3 + 7x^2 + 15x + 9$

$= (-1)^3 + 7(-1)^2 + 15(-1) + 9$

$= -1 + 7 - 15 + 9$

$= 0$

Let : $f(x) = x^2 + 4x + 3$

$g(x) = x + 3$

Find: $\left(\frac{f}{g}\right)(3)$

$= (x^2 + 4x + 3)/(x + 3)$

$= (x + 3)(x + 1)/(x + 3)$

$= x + 1$

$= 3 + 1$

$= 4$

Let : $f(x) = x^2 + 10x - 11$

$g(x) = x^2 + 22$

Find the following:

$(f + g)(3)$

$(f - g)(1)$

$(f \bullet g)(4)$

$\left(\frac{f}{g}\right)(-1)$

Composite Function

Composite Function

The output of one function serves as the input of another function

$$(f \circ g)(x) \quad = f[\, g(x) \,]$$

Let : $f(x) = x^2 - 17$

$g(x) = x + 3$

Find: $(f \circ g)(x)$

$= f[\, g(x)\,]$

$= f(\, x + 3\,)$; $f(x) = x^2 - 17$

$= (x + 3)^2 - 17$

$= x^2 + 6x + 9 - 17$

$= x^2 + 6x - 8$

Let : $f(x) = x^2 - x$

$g(x) = x + 5$

Find: $(f \circ g)(1)$

$= f[\, g(x)\,]$

$= f(\, x + 5\,)$; $f(x) = x^2 - x$

$= (x + 5)^2 - (x + 5)$

$= x^2 + 10x + 25 - x - 5$

$= x^2 + 9x + 20$

$= 1^2 + 9(1) + 20$

$= 30$

Let : $f(x) = \mathrm{x}^2 - \mathrm{x}$

$g(x) = \mathrm{x} + 5$

Find: $(g \circ f)(2)$

$= g\,[f(x)\,]$

$= g(\mathrm{x}^2 - \mathrm{x}\,)$; $g(x) = \mathrm{x} + 5$

$= (\mathrm{x}^2 - \mathrm{x}\,) + 5$

$= \mathrm{x}^2 - \mathrm{x} + 5$

$= 2^2 - 2 + 5$

$= 7$

Let : $f(x) = x^2 - x$

$g(x) = x + 5$

Find: $(f \circ f)(-1)$

$= f[f(x)]$

$= f(x^2 - x)$; $f(x) = x^2 - x$

$= (x^2 - x)^2 - (x^2 - x)$

$= x^4 - 2x^3 + x^2 - x^2 + x$

$= x^4 - 2x^3 + x$

$= (-1)^4 - 2(-1)^3 + (-1)$

$= 1 + 2 - 1$

$= 2$

Let : $f(x) = \mathrm{x}^2 - \mathrm{x}$

$g(x) = \mathrm{x} + 5$

Find: $(g \circ g)(0)$

$= g\,[g(x)\,]$

$= g(\,\mathrm{x} + 5\,)$; $g(x) = \mathrm{x} + 5$

$= (\,\mathrm{x} + 5\,) + 5$

$= \mathrm{x} + 10$

$= 0 + 5$

$= 5$